Data Cleansing & Quality Audit Log

Project: E-Commerce Customer Retention Intelligence

Scope: Registered Customer Cohort Analysis

Status: Verified & Reconciled

1. Executive Summary & Data Integrity Rules

This audit log documents the data cleaning, transformation, and quality assurance processes applied to the raw e-commerce transaction dataset prior to semantic modeling in Power BI. To satisfy the Business Requirements Document (BRD), the core data processing strategy isolates active registered customer behavior, explicitly separating non-qualifying and guest checkout transactions.

> **Primary Integrity Objective:** Establish a clean, verified registered baseline of **9,607 unique customers** generating **$3,108,791.35 in net sales** for accurate RFM scoring and retention strategy execution.

2. Cleansing Steps & Filtering Logic

Step 1: Order Status Filtering

Completed Only

- **Rule:** Filter strictly for `OrderStatus = "Completed"`.
- **Rationale:** Excludes cancelled, pending, failed, or refunded transactions to ensure monetary calculations reflect true realized revenue.

Step 2: Customer Identity Isolation (Guest Checkout Exclusion)

Cohort Isolation

- **Rule:** Exclude rows where `CustomerID` is null, blank, or anonymous (`CustomerID is not blank`).
- **Guest Checkouts Excluded:** 1,407 orders representing **$101,140.00** in net sales.
- **Registered Cohort Retained: 9,607 unique registered customers** generating **$3,108,791.35** across 18,214 completed orders.

Step 3: Date Table Engineering (dim_date)

Calendar Master

- **Rule:** Built a dedicated, continuous calendar dimension table (`dim_date`) spanning `2023-01-01` to `2025-12-31`.
- **Attributes Generated:** DateKey, Year, Quarter, Month, MonthName, DayOfWeek, FiscalYear, IsWeekend.
- **Relationship:** Joined `dim_date[Date]` (1) to `fact_orders[OrderDate]` (*) to enable time-intelligence calculations and cohort comparisons.

Step 4: Granularity & Order-Grain Verification

Order Grain

- **Rule:** Enforce order frequency at distinct order grain using `DISTINCTCOUNT(fact_orders[OrderID])`.
- **Rationale:** Prevents multi-item line orders from artificially inflating customer purchase frequency scores.

3. Reconciled Baseline Metrics

The table below provides a full reconciliation between raw imported transactional tables and the final cleansed registered customer cohort model:

Metric Name	Raw Dataset	Cleansed Registered Cohort	Variance / Impact Analysis
Total Customer Accounts	11,014	**9,607**	-1,407 Guest Checkout Accounts Excluded
Completed Orders	19,621	**18,214**	-1,407 Non-Registered Orders Excluded
Total Net Sales Revenue	$3,209,931.35	**$3,108,791.35**	-$101,140.00 Non-Registered Revenue Excluded
Avg Spend per Registered User	N/A	**$323.60**	Baseline established for RFM cohort scoring

4. Sign-Off & Verification Checks

Data Quality Checkpoint Results **PASSED ALL CHECKS**

- **Primary Key Integrity:** 100% of `CustomerID` keys in `rfm_customer_scores.csv` are valid and mapped to stable customer master keys.
- **RFM Scoring Engine Validation:** Recency (1–5), Frequency (1–5), and Monetary (1–5) scores successfully calculated for all 9,607 registered accounts without null or missing values.
- **Temporal Alignment:** All transactional order dates cleanly join against the continuous `dim_date` table spanning 2023–2025.

5. Next Steps & Downstream Consumption

The cleansed dataset and `dim_date` dimension are fully verified. All downstream DAX measures (Base Measures, RFM Scoring Rules, and Segment Classifications) consume these cleansed layers to feed the 3-page Power BI Customer Retention Dashboard.